

Predicting Crime Severity from NYPD Incident Data

A Multiclass Classification Analysis

2.5 Million Incident Records

Objective: Crime Severity Classification

Analyze NYPD incident-level data to classify crime severity. The key objective is **not to claim strong predictive performance**, but to show how EDA and feature selection set realistic expectations, justify modeling choices, and clearly explain what this workflow can and cannot deliver.

Workflow Stages

1. **EDA Insights:** What information differentiates severity?
2. **Feature Selection:** Which variables carry signal?
3. **Modeling & Results:** Can we extract that signal?
4. **Limitations & Recommendations:** What would actually improve results?

Data Overview & Features Dictionary

Dataset: 2.5M+ NYPD incident records

Features: Temporal, spatial, demographic, offense-level

Target: Severity (Violation / Misdemeanor / Felony)

Feature	Description	Type
PREM_GROUPED	Premises grouping (engineered)	Categorical
RACE_x_BORO	Race × Borough interaction	Categorical
VIC_SEX_MAPPED	Victim sex (cleaned)	Categorical
VIC_AGE_GROUP	Victim age group	Categorical
VIC_RACE_MAPPED	Victim race (cleaned)	Categorical
SUSP_SEX_MAPPED	Suspect sex (cleaned)	Categorical
SUSP_RACE_MAPPED	Suspect race (cleaned)	Categorical
Latitude / Longitude	Geographic coordinates (NYC bounds)	Numeric
Report_Delay_Hours	Hours between incident and report	Numeric
hour_x_delay	Hour × Report_Delay interaction	Numeric
Severity	Target: Violation (0), Misdemeanor (1), Felony (2)	Target

Dataset & Data Cleaning

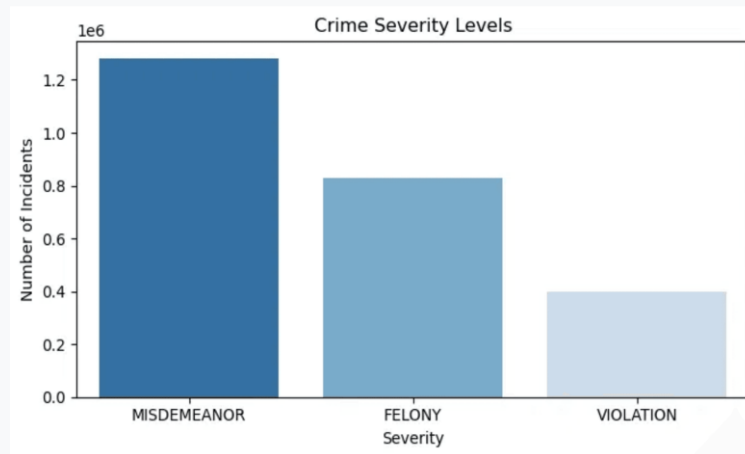
Data Cleaning Steps

1 Removal of identifier and administrative columns

2 Dropping columns with excessive missingness

3 Standardizing categorical values

4 Filtering inconsistent or invalid records



Severity Mapping

LAW_CAT_CD Mapping

Violation	Level 0
Misdemeanor	Level 1
Felony	Level 2

Feature Engineering

01

Premises Grouping

Consolidated over 200 raw premises categories (PREM_TYP_DESC) into 10 high-level groups to reduce dimensionality and capture broader spatial patterns.

02

Race × Borough Interaction

Created a demographic-geographic interaction feature to capture localized patterns of crime severity across different NYC boroughs.

03

Report Delay Hours

Calculated the precise time difference between the incident occurrence and the formal police report to identify reporting behaviors linked to severity.

04

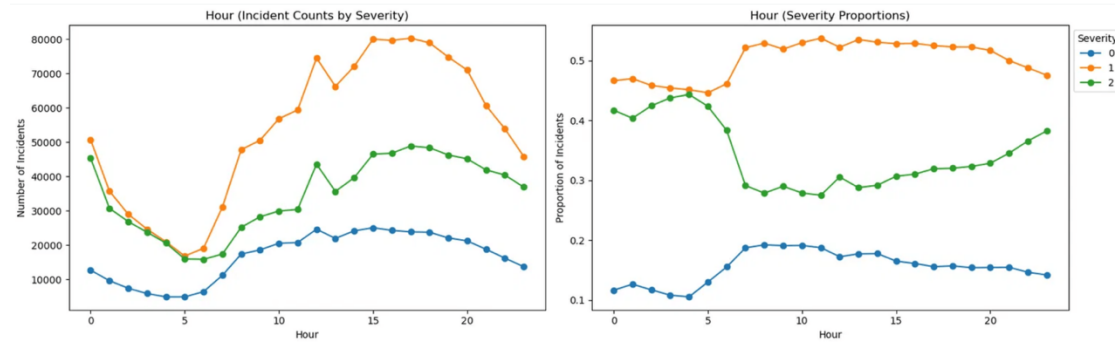
Hour × Delay Interaction

Combined the time of day with the report delay to capture complex temporal signals that might indicate the urgency or nature of the incident.

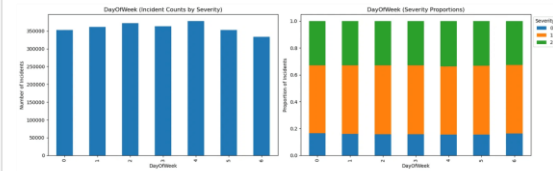
EDA: Temporal Features

Temporal patterns reveal differential severity distribution across time periods.

Hour of Day (Primary Signal)

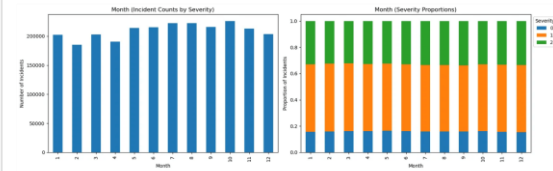


Day of Week



Limited signal

Month



No signal

Hour-of-day carries severity signal; day-of-week and month do not meaningfully differentiate severity.

EDA: Demographic Features

Victim vs. Suspect

Victim features consistently showed slightly stronger association with severity than suspect features.

Effect Size

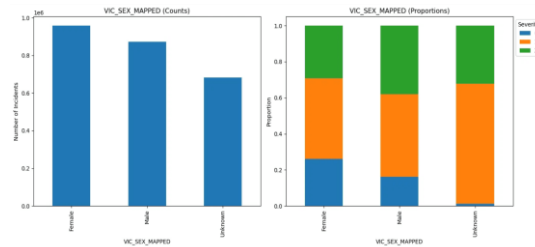
Both sets of variables showed dependence with severity, but effect sizes were small.

Data Completeness

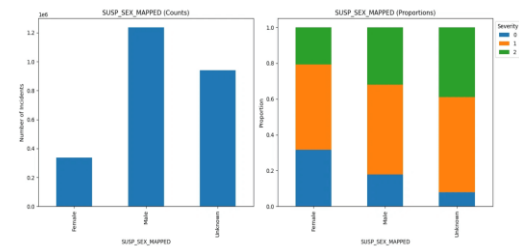
Suspect demographics were often missing or incomplete, reducing their usefulness.

EDA Takeaway: Victim characteristics marginally more informative than suspect characteristics, but neither provides strong separation on its own.

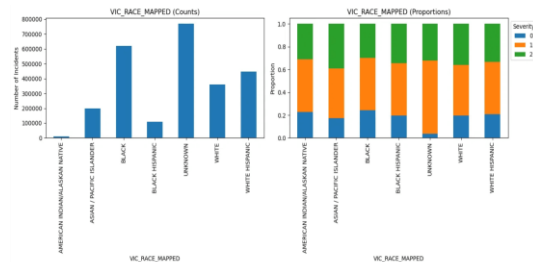
Victim Sex



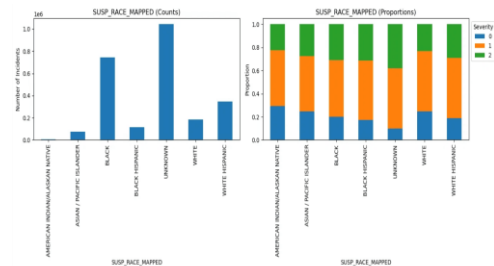
Suspect Sex



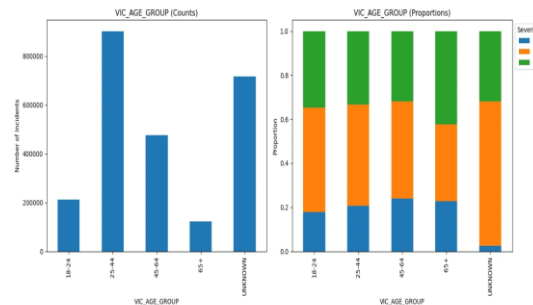
Victim Race



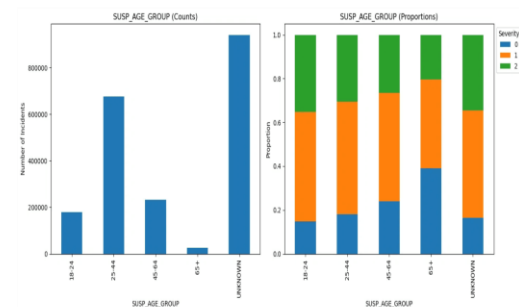
Suspect Race



Victim Age



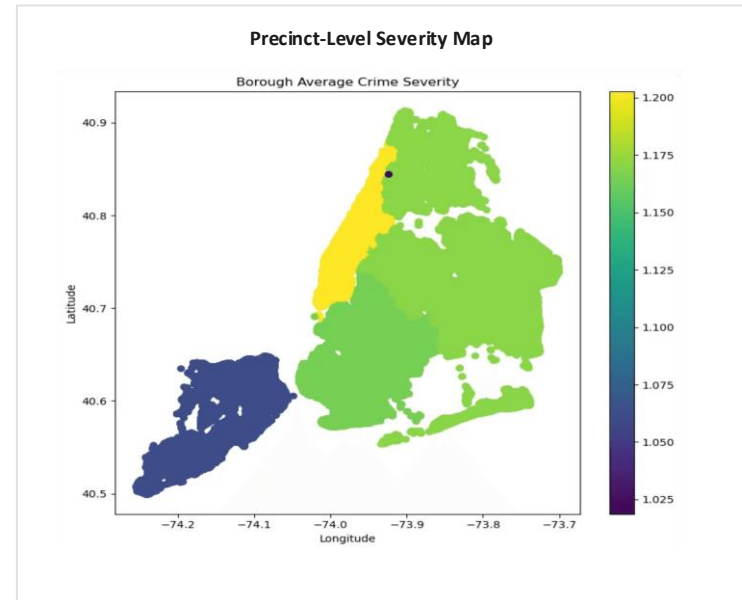
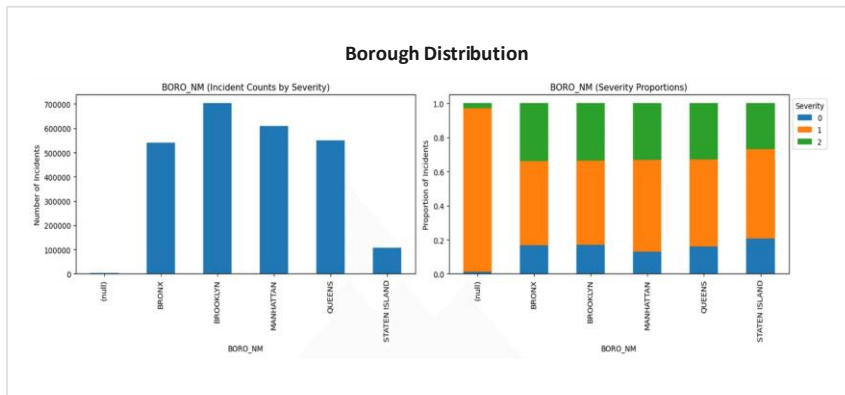
Suspect Age



EDA: Spatial Features

Spatial variables such as borough and precinct showed moderate association with severity.

Certain areas exhibit higher proportions of severe crimes, but overlap between severity classes across locations is substantial. Location matters, but mostly at an aggregate level; it does not uniquely determine severity.



Feature Selection: Statistical Tests & Effect Sizes

Chi-Squared Test: All 26 features show statistical significance ($p < 0.05$).

Cramér's V: Measures effect size to identify features with practical predictive value.

Retained Features ($V > 0.10$)

PREM_TYP_DESC	0.215
RACE_x_BORO	0.210
VIC_SEX_MAPPED	0.204
PREM_GROUPED	0.200
VIC_AGE_GROUP	0.187
VIC_RACE_MAPPED	0.176
SUSP_SEX_MAPPED	0.155
SUSP_RACE_MAPPED	0.114

Dropped Features (Weak or Leakage)

PD_DESC	0.999	LEAKAGE
OFNS_DESC	0.958	LEAKAGE
LOC_OF_OCCUR_DESC	0.096	
ADDR_PCT_CD	0.077	
HOUSING_PSA	0.064	
BORO_NM	0.047	

Critical Finding:
Retained features show weak associations (max $V = 0.215$). This explains why all models converge on ~55% accuracy—the feature signal itself is limited.

Feature Relationships: Cramér's V Analysis

Statistical Measure

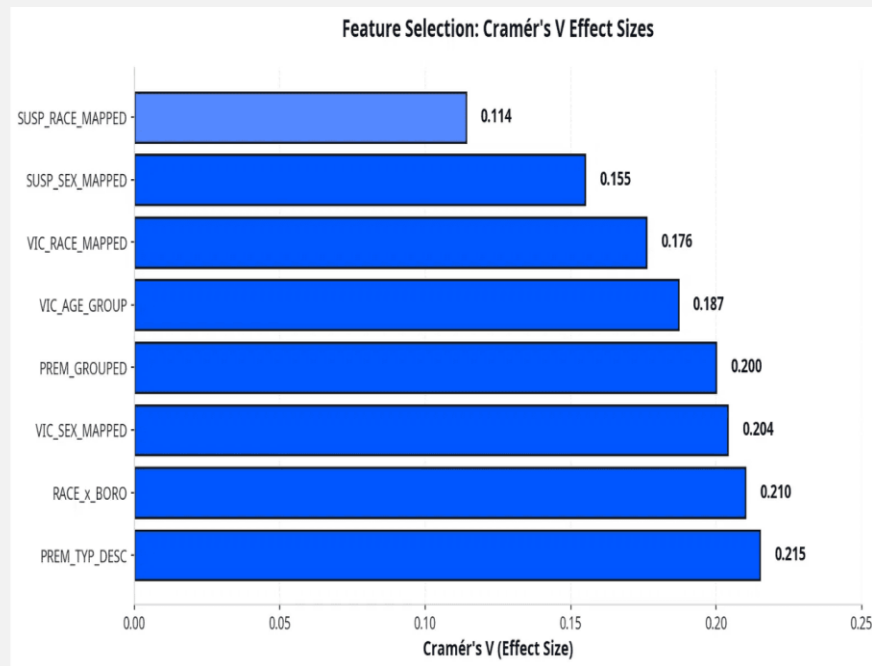
Cramér's V measures strength of association between categorical variables on a scale from 0 to 1.

Key Finding

Strongest Cramér's V values were around 0.3, with most well below that. This indicates weak-to-moderate associations.

Performance Ceiling

This result alone suggested—before modeling—that performance would be limited. Models cannot exceed the information content of their features.



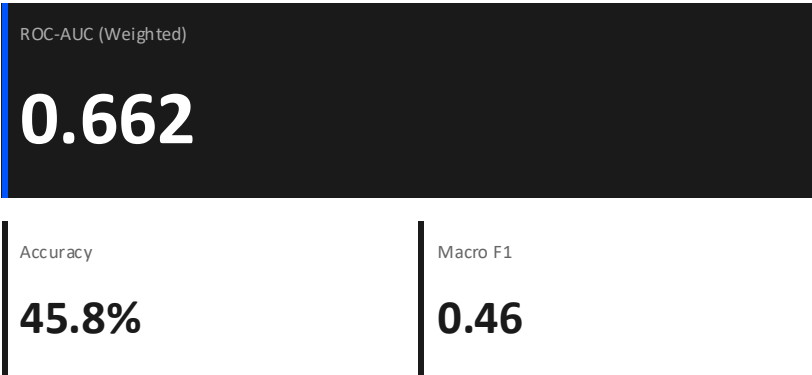
Model 1: Logistic Regression

Multinomial Logistic Regression (Baseline)

Model Configuration

Class Weight	balanced
Max Iterations	1000
Multi-class	multinomial
Random State	42

Performance (Test Set)



Key Insight: Linear model confirms that severity is not linearly separable using available features. Baseline establishes a performance floor.

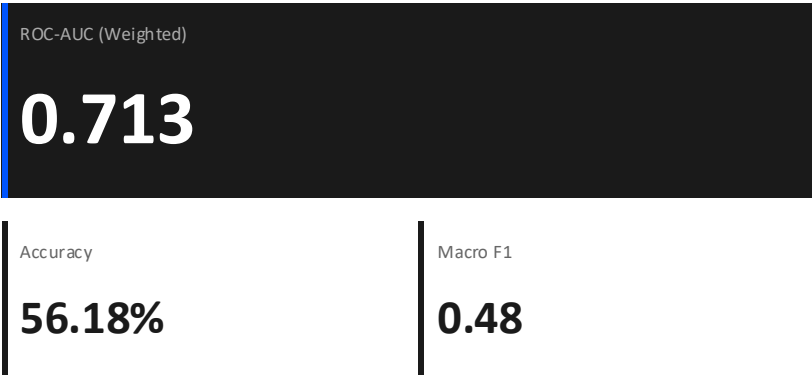
Model 2: XGBoost

Extreme Gradient Boosting (Tuned)

Best Parameters (Grid Search)

Learning Rate	0.05
Max Depth	8
N Estimators	500
Subsample	0.8

Performance (Test Set)



Key Insight: Best performer among all models. Non-linear interactions and feature importance weighting improve results, but gains remain incremental. Confirms feature signal is the limiting factor.

Model 3: XGBoost with Random Oversampling

Strategy & Goal

Sampling Strategy

Applied **RandomOverSampler** to the training data to balance class distributions, specifically targeting minority classes.

Primary Objective

Improve **recall** for high-severity incidents (Felony), which are often overshadowed by the majority class.

Impact on Resource Allocation

ROS significantly improves recall for Felonies and Violations.

This shift is critical for law enforcement resource allocation, prioritizing the detection of high-severity incidents over raw accuracy.

While overall accuracy may decrease, the model becomes more effective at identifying incidents that require **immediate and high-level response**.

STRATEGIC SHIFT:

Prioritizing Severity over Accuracy

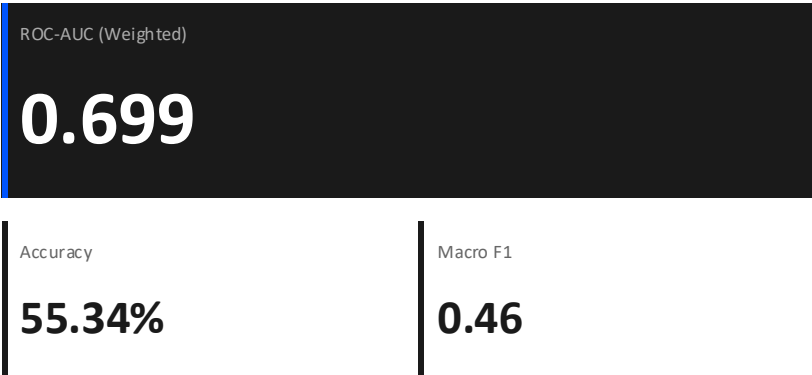
Model 3: Neural Network

Multilayer Perceptron (Tuned)

Best Parameters (Grid Search)

Hidden Layers	(150, 100)
Activation	relu
Solver	adam
Alpha (L2)	0.01

Performance (Test Set)



Key Insight: Despite extensive tuning, the neural network underperforms XGBoost. Added complexity does not translate to better generalization, suggesting the primary constraint is feature signal, not model flexibility.

Model Comparison

Test Set Performance Across All Three Models

Model	ROC-AUC (Weighted)	Accuracy	Macro F1	Felony Recall
Logistic Regression	0.658	44.96%	0.45	0.54
XGBoost	0.713	56.18%	0.48	0.31
XGBoost with ROS	0.686	49.62%	0.49	0.52
Neural Network	0.699	55.34%	0.46	0.33

Key Finding:

This comparison shows that all models achieve similar performance, with ROC-AUC between 0.66 and 0.71, indicating that the problem is largely data-limited rather than model-limited. XGBoost without oversampling performs best overall, achieving the highest ROC-AUC and accuracy while maintaining a reasonable macro F1-score. Although oversampling improves felony recall, it significantly reduces precision and overall accuracy across models.

Limitations

Category 01

Data Constraints

- No prior criminal history or escalation patterns available.
- Missing explicit victim–suspect relationship data.
- Absence of narrative context or direct officer observations.

Category 02

Modeling Constraints

- Limited time window for exhaustive hyperparameter tuning.
- Hardware constraints on training complex ensemble models.
- Restricted exploration of advanced deep learning architectures.

Category 03

Statistical Ceiling

- Feature associations are inherently weak ($V \leq 0.3$).
- Models cannot exceed the information content of features.
- Performance plateau is a predictable result of data signal.

Key Takeaway

Significant improvements require both **better data** (more context and history) and **more computational resources** (time and hardware for advanced tuning). Neither alone is sufficient.

Operational Implications

Appropriate Uses

Post-Incident Analysis

Identifying patterns and trends across precincts to inform long-term strategy.

Resource Planning

Aggregate severity trend forecasting for departmental budgeting and staffing.

Training Tool

Use the model to point out cases where it disagrees with the officer, so a human can take a second look

Inappropriate Uses

Real-Time Prediction

Immediate response dispatch without human review or situational context.

Automated Classification

Replacing officer judgment with algorithmic outputs for legal reporting.

Technological Reliance

Assuming model tuning can resolve fundamental gaps in incident data.

Strategic insight, not tactical automation.

Conclusion

Better Information, Not Better Algorithms

EDA revealed that weak feature associations constrain all models. Improvements require better data and computational resources, not algorithmic innovation.

This is a data strategy problem, not an algorithm problem.